Delays adversarial

1 Model

Fix a horizon $T \in \mathbb{N}$. Before the game begins, an oblivious adversary chooses a deterministic sequence

$$p = (p_1, \dots, p_T) \in [0,1]^T.$$

At every round $t = 1, \dots, T$:

(1) the learner observes p_t;

(2) the learner posts a possibly randomized quote $A_t \in [0,1]$ based only on $p_1, \dots, p_t$ and its internal randomization.

The learner's actions do not affect the price sequence.

For a deterministic quote $a \in [0,1]$ posted at time t, define its execution delay by

$$\tau_t^p(a) := \min\bigl\{d \in \{1, \dots, T-t\} : p_{t+d} > a\bigr\},$$

with the convention $\min \varnothing = \infty$. Its reward is defined as

$$r_t^p(a) := a\, \mathbf{1}\{\tau_t^p(a) < \infty\}\,.$$

Thus, the order executes at the first strictly larger future price and earns its posted quote.

A randomized policy π is a collection of nonanticipating decision rules for $A_1, \dots, A_T$. Its regret on p is

$$R_T(\pi, p) := \sup_{a \in [0,1]} \sum_{t=1}^{T} r_t^p(a) - \mathbb{E}_\pi\left[\sum_{t=1}^{T} r_t^p(A_t)\right], \tag{1}$$

where the expectation is only over the learner's randomization.

2 Reward-weighted delay tails

For an integer truncation level $H \geq 0$, define the truncated reward

$$r_{t,H}^p(a) := a\, \mathbf{1}\{\tau_t^p(a) \leq H\}\,.$$

The reward lost by truncating quote a after H rounds is therefore

$$\sum_{t=1}^{T} \bigl(r_t^p(a) - r_{t,H}^p(a)\bigr) = \sum_{t=1}^{T} a\, \mathbf{1}\{H < \tau_t^p(a) < \infty\}\,.$$

This motivates the cumulative reward-weighted tail

$$\Gamma_{T,H}(p) := \sup_{a\in[0,1]} \sum_{t=1}^{T} a\,\mathbf{1}\{H < \tau_t^p(a) < \infty\}\,. \tag{2}$$

It measures the largest total reward that a single fixed quote obtains from orders whose execution delay is strictly larger than H.

Let

$$g : \mathbb{N}_0 \longrightarrow [0,1]$$

be nonincreasing, where $\mathbb{N}_0 = \{0,1,2,\ldots\}$. For each horizon T, define the maximal tail-envelope class of price sequences

$$\mathcal{C}_T(g) := \left\{p \in [0,1]^T : \Gamma_{T,H}(p) \le T g(H) \text{ for every } H \in \mathbb{N}_0\right\}. \tag{3}$$

We restrict attention to adversaries that select a sequence from this class. The learner knows T and g, but not the sequence selected from $\mathcal{C}_T(g)$. Define the minimax regret

$$\mathfrak{R}_T(g) := \inf_{\pi} \sup_{p\in\mathcal{C}_T(g)} R_T(\pi,p). \tag{4}$$

Examples of tail envelopes

Every price sequence satisfies $\Gamma_{T,H}(p) \le T$, and hence belongs to $\mathcal{C}_T(g)$ for the uninformative envelope $g \equiv 1$. The following examples give more informative envelopes.

Uniformly fast execution. Suppose that, for some $D \ge 1$, every order that eventually executes does so within D rounds:

$$\tau_t^p(a) < \infty \quad\Longrightarrow\quad \tau_t^p(a) \le D \qquad \text{for every } t \text{ and } a.$$

Then $\Gamma_{T,H}(p) = 0$ for $H \ge D$, so $p \in \mathcal{C}_T(g_D)$ for $g_D(H) := \mathbf{1}\{H < D\}$. In particular, every nonincreasing price sequence has this property with $D = 1$: if some later price exceeds a, then already $p_{t+1} > a$.

Recurring spikes. Let $p_t \in \{0,1\}$, assume $p_1 = 1$, and write $1 = s_1 < s_2 < \cdots < s_m$ for the times at which $p_t = 1$. If $d_i := s_{i+1} - s_i$, then

$$\Gamma_{T,H}(p) = \sum_{i=1}^{m-1} (d_i - H)_+, \qquad (u)_+ := \max\{u,0\}.$$

Indeed, within a gap of length d_i, exactly $(d_i - H)_+$ orders wait more than H rounds for the next spike; the supremum in (2) is approached as $a \uparrow 1$. Thus regularly spaced spikes with gap D satisfy

$$\Gamma_{T,H}(p) \le T\left(1 - \frac{H}{D}\right)_+,$$

so one may take $g(H) = (1 - H/D)_+$. More generally, long gaps are allowed provided that they occupy a suitably small fraction of time.

A single terminal jump. For $p = (0, \ldots, 0, x)$ with $x \in (0, 1]$,

$$\Gamma_{T,H}(p) = x(T - H - 1)_+.$$

Consequently, this sequence belongs to $\mathcal{C}_T(g_x)$ for the constant envelope $g_x(H) \equiv x$. Although any one finite path has zero tail once $H \geq T - 1$, the family of terminal-jump paths across all horizons cannot be covered by a single envelope that tends to zero.

3 Main characterization

Theorem 1 (Adversarial delay-tail characterization)**.** *Let $g : \mathbb{N}_0 \to [0, 1]$ be nonincreasing, and let $\mathfrak{R}_T(g)$ be defined by* (4)*.*

(i) For every $T \in \mathbb{N}$, every integer $H \geq 0$, and every integer $K \geq 2$,

$$\mathfrak{R}_T(g) \leq \sqrt{\frac{T(H+1)\log K}{2}} + \frac{T}{K} + Tg(H). \tag{5}$$

(ii) Let

$$c := \lim_{H \to \infty} g(H),$$

which exists because g is nonincreasing. If $c > 0$, then for every $T \in \mathbb{N}$,

$$\mathfrak{R}_T(g) \geq \frac{c}{e}(T - 1). \tag{6}$$

Consequently,

$$\boxed{\mathfrak{R}_T(g) = o(T) \quad \Longleftrightarrow \quad \lim_{H \to \infty} g(H) = 0.} \tag{7}$$

Why the theorem is true

If $g(H)$ is small, then no fixed quote can obtain much total reward from orders that remain unresolved for more than H rounds. We may therefore truncate all orders after H rounds at a cost of at most $Tg(H)$ against the best fixed comparator. After truncation, the complete reward vector for round t is known by time $t + H$. Splitting time into $H + 1$ residue classes eliminates this delay: before the next decision in a residue class, the preceding reward vector in that class is already known. Hedge can then be run separately on each class. Finally, rounding every comparator quote downward to a grid loses at most $1/K$ per round.

For the converse, suppose $g(H)$ remains bounded below by $c > 0$. The class then contains every terminal-jump sequence

$$(0, \ldots, 0, x), \qquad 0 \leq x \leq c.$$

Before the final round, all such sequences have exactly the same observed prefix, so the learner must post its first $T - 1$ quotes without knowing x. A suitable distribution over terminal values creates an unavoidable loss of c/e per preterminal round relative to the best fixed quote in hindsight.

The proof is divided into an upper-bound part and a lower-bound part.

4 Upper bound

4.1 Truncation

Lemma 2 (Pathwise truncation comparison)**.** *Fix T, H, a sequence $p \in [0,1]^T$, and a policy π. Define the truncated regret*

$$R_{T,H}(\pi, p) := \sup_{a \in [0,1]} \sum_{t=1}^{T} r^p_{t,H}(a) - \mathbb{E}_\pi \left[\sum_{t=1}^{T} r^p_{t,H}(A_t) \right].$$

Then

$$R_T(\pi, p) \le R_{T,H}(\pi, p) + \Gamma_{T,H}(p). \tag{8}$$

Proof. For every fixed $a \in [0,1]$, definition (2) gives

$$\sum_{t=1}^{T} r^p_t(a) \le \sum_{t=1}^{T} r^p_{t,H}(a) + \Gamma_{T,H}(p).$$

Taking the supremum over a yields

$$\sup_a \sum_{t=1}^{T} r^p_t(a) \le \sup_a \sum_{t=1}^{T} r^p_{t,H}(a) + \Gamma_{T,H}(p).$$

Moreover, $r^p_t(a) \ge r^p_{t,H}(a)$ for every t, a, so pathwise

$$\sum_{t=1}^{T} r^p_t(A_t) \ge \sum_{t=1}^{T} r^p_{t,H}(A_t).$$

Taking expectations and combining the two inequalities proves (8). □

4.2 Discretization

For $K \ge 2$, let

$$\mathcal{G}_K := \left\{ 0, \frac{1}{K}, \ldots, \frac{K-1}{K} \right\}. \tag{9}$$

Lemma 3 (Downward discretization)**.** *Fix T, H, p, and suppose that the learner always chooses $A_t \in \mathcal{G}_K$. Define its grid-comparator truncated regret by*

$$R_{T,H,K}(\pi, p) := \max_{b \in \mathcal{G}_K} \sum_{t=1}^{T} r^p_{t,H}(b) - \mathbb{E}_\pi \left[\sum_{t=1}^{T} r^p_{t,H}(A_t) \right].$$

Then

$$R_{T,H}(\pi, p) \le R_{T,H,K}(\pi, p) + \frac{T}{K}. \tag{10}$$

Proof. For $a \in [0,1]$, define its downward grid approximation

$$b(a) := \frac{\min\{\lfloor Ka \rfloor, K-1\}}{K}.$$

Then $b(a) \in \mathcal{G}_K$, $b(a) \le a$, and

$$0 \le a - b(a) \le \frac{1}{K}.$$

Fix a and write $b = b(a)$. Since $b \le a$, every future price that exceeds a also exceeds b. Hence

$$\mathbf{1}\{\tau_t^p(a) \le H\} \le \mathbf{1}\{\tau_t^p(b) \le H\}.$$

Therefore, pathwise,

$$\begin{aligned} r_{t,H}^p(a) &= a\mathbf{1}\{\tau_t^p(a) \le H\} \\ &= b\mathbf{1}\{\tau_t^p(a) \le H\} + (a-b)\mathbf{1}\{\tau_t^p(a) \le H\} \\ &\le b\mathbf{1}\{\tau_t^p(b) \le H\} + \frac{1}{K} \\ &= r_{t,H}^p(b) + \frac{1}{K}. \end{aligned}$$

Summing over t and taking the supremum over a gives

$$\sup_{a\in[0,1]} \sum_{t=1}^{T} r_{t,H}^p(a) \le \max_{b\in\mathcal{G}_K} \sum_{t=1}^{T} r_{t,H}^p(b) + \frac{T}{K}.$$

Subtracting the learner's expected truncated reward proves (10). □

4.3 A self-contained Hedge bound

We first record Hoeffding's lemma in the form needed below.

Lemma 4 (Hoeffding's lemma)**.** *If $X \in [0,1]$ almost surely, then for every $\eta \in \mathbb{R}$,*

$$\log \mathbb{E}[e^{\eta X}] \le \eta\mathbb{E}[X] + \frac{\eta^2}{8}.$$

Proof. Let

$$\phi(\eta) := \log \mathbb{E}[e^{\eta X}] - \eta\mathbb{E}[X].$$

Under the exponentially tilted distribution proportional to $e^{\eta X}$, $\phi''(\eta)$ is the variance of X. Every random variable supported on $[0,1]$ has variance at most $1/4$, so $\phi''(\eta) \le 1/4$. Also $\phi(0) = \phi'(0) = 0$. Integrating the second-derivative bound twice gives $\phi(\eta) \le \eta^2/8$. □

Lemma 5 (Hedge with full-information gains)**.** *Let $x_1, \dots, x_n \in [0,1]^K$ be arbitrary gain vectors. There is a randomized online algorithm which chooses $I_s \in \{1, \dots, K\}$ before seeing x_s, observes all of x_s afterward, and satisfies*

$$\max_{i\in\{1,\dots,K\}} \sum_{s=1}^{n} x_{s,i} - \mathbb{E}\left[\sum_{s=1}^{n} x_{s,I_s}\right] \le \sqrt{\frac{n \log K}{2}}. \tag{11}$$

For $n = 0$, the regret is zero.

Proof. Assume $n \ge 1$. Initialize $w_{1,i} = 1$ for every i. At local round s, play arm i with probability

$$q_{s,i} = \frac{w_{s,i}}{\sum_j w_{s,j}},$$

and, after observing x_s, update

$$w_{s+1,i} = w_{s,i} e^{\eta x_{s,i}}.$$

Let $W_s = \sum_i w_{s,i}$. By Lemma 4, applied to a random variable that equals $x_{s,i}$ with probability $q_{s,i}$,

$$\log \frac{W_{s+1}}{W_s} = \log \sum_i q_{s,i} e^{\eta x_{s,i}} \le \eta \sum_i q_{s,i} x_{s,i} + \frac{\eta^2}{8}.$$

Summing over s gives

$$\log \frac{W_{n+1}}{W_1} \le \eta \sum_{s=1}^{n} \sum_i q_{s,i} x_{s,i} + \frac{n\eta^2}{8}.$$

For every fixed arm i,

$$W_{n+1} \ge w_{n+1,i} = \exp\left(\eta \sum_{s=1}^{n} x_{s,i}\right),$$

while $W_1 = K$. Hence

$$\sum_{s=1}^{n} x_{s,i} - \sum_{s=1}^{n} \sum_j q_{s,j} x_{s,j} \le \frac{\log K}{\eta} + \frac{n\eta}{8}.$$

The expected gain of the randomized algorithm at round s is $\sum_j q_{s,j} x_{s,j}$. Maximizing over i and choosing

$$\eta = \sqrt{\frac{8 \log K}{n}}$$

proves (11). □

4.4 Learning the truncated problem

Lemma 6 (Interlaced Hedge)**.** *For every T, H, K, there exists a nonanticipating grid-valued policy $\pi_{H,K}$ such that, for every deterministic sequence $p \in [0,1]^T$,*

$$R_{T,H,K}(\pi_{H,K}, p) \le \sqrt{\frac{T(H+1)\log K}{2}}. \tag{12}$$

Proof. Partition the rounds into $H+1$ residue classes modulo $H+1$:

$$S_j := \{t \in \{1,\dots,T\} : t \equiv j \pmod{H+1}\}, \qquad j = 1,\dots,H+1,$$

using any consistent convention for residues. Run one independent Hedge instance on each S_j, with arms indexed by the quotes in $\mathcal{G}_K$.

For a round t, define the full-information gain vector

$$x_t(b) := r^p_{t,H}(b), \qquad b \in \mathcal{G}_K.$$

This vector is completely determined after observing

$$p_{t+1}, \dots, p_{\min\{t+H,T\}}.$$

If t' is the next round after t in the same residue class, then $t' = t + H + 1$. Thus x_t is known no later than round $t' - 1$, before the next decision of the same Hedge instance. Consequently, each residue-class instance receives ordinary immediate full-information feedback on its own local time scale.

Let $n_j = |S_j|$. Lemma 5 gives local expected regret at most $\sqrt{n_j \log K/2}$. Since

$$\max_{b\in\mathcal{G}_K} \sum_{t=1}^{T} x_t(b) \le \sum_{j=1}^{H+1} \max_{b\in\mathcal{G}_K} \sum_{t\in S_j} x_t(b),$$

the global grid-comparator regret is at most the sum of the local regrets. By Cauchy–Schwarz,

$$\begin{aligned} R_{T,H,K}(\pi_{H,K}, p) &\le \sum_{j=1}^{H+1} \sqrt{\frac{n_j \log K}{2}} \\ &\le \sqrt{(H+1)\left(\sum_{j=1}^{H+1} n_j\right)\frac{\log K}{2}} \\ &= \sqrt{\frac{T(H+1)\log K}{2}}. \end{aligned}$$

□

4.5 Proof of the upper bound

Proof of Theorem 1(i). Fix $H \ge 0$ and $K \ge 2$, and use the policy $\pi_{H,K}$ from Lemma 6. For every $p \in \mathcal{C}_T(g)$, Lemmas 2, 3, and 6 give

$$\begin{aligned} R_T(\pi_{H,K}, p) &\le R_{T,H}(\pi_{H,K}, p) + \Gamma_{T,H}(p) \\ &\le R_{T,H,K}(\pi_{H,K}, p) + \frac{T}{K} + \Gamma_{T,H}(p) \\ &\le \sqrt{\frac{T(H+1)\log K}{2}} + \frac{T}{K} + Tg(H). \end{aligned}$$

The right-hand side is uniform over $p \in \mathcal{C}_T(g)$. Taking the supremum over p and then the infimum over all policies proves (5). □

5 Lower bound

The lower bound uses sequences whose only positive price occurs at the terminal round.

Lemma 7 (Tail mass of a terminal jump)**.** *For $x \in [0,1]$, let*

$$p^x = (0, \ldots, 0, x) \in [0,1]^T.$$

Then, for every integer $H \ge 0$,

$$\Gamma_{T,H}(p^x) = x\,(T - H - 1)_+, \tag{13}$$

where $(z)_+ = \max\{z, 0\}$.

Proof. Fix a quote a. If $a \ge x$, it never executes. If $a < x$, then for every $t < T$ its first execution occurs at time T, so

$$\tau_t^{p^x}(a) = T - t.$$

The delay is strictly larger than H exactly for $t = 1, \dots, T - H - 1$, when this set is nonempty. Hence the late-trade reward of $a < x$ is

$$a\,(T - H - 1)_+.$$

Taking the supremum over $a < x$ gives $x(T - H - 1)_+$. The value need not be attained because a quote equal to x does not execute. $\square$

Lemma 8 (Membership of terminal jumps)**.** *Let* $c = \lim_{H\to\infty} g(H) > 0$. *Then*

$$p^x \in \mathcal{C}_T(g) \qquad \textit{for every } x \in [0, c].$$

Proof. Because g is nonincreasing with limit c, we have $g(H) \ge c$ for every H. By Lemma 7, for $x \le c$,

$$\Gamma_{T,H}(p^x) = x(T - H - 1)_+ \le xT \le cT \le Tg(H).$$

This is precisely the defining condition of $\mathcal{C}_T(g)$. $\square$

Lemma 9 (Hard distribution over terminal prices)**.** *Fix* $c \in (0, 1]$. *There exists a random variable* $X \in [0, c]$ *such that*

$$\mathbb{E}[X] = \frac{2c}{e} \qquad \textit{and} \qquad a \Pr(X > a) \le \frac{c}{e} \quad \textit{for every } a \in [0, 1]. \tag{14}$$

Proof. Define X through its survival function

$$\Pr(X > x) = \begin{cases} 1, & 0 \le x < c/e, \\ \dfrac{c}{ex}, & c/e \le x < c, \\ 0, & x \ge c. \end{cases}$$

Equivalently, X has density $c/(ex^2)$ on $[c/e, c)$ and an atom of mass $1/e$ at c. Using the tail-integral formula,

$$\begin{aligned} \mathbb{E}[X] &= \int_0^c \Pr(X > x)\,dx \\ &= \frac{c}{e} + \int_{c/e}^{c} \frac{c}{ex}\,dx \\ &= \frac{c}{e} + \frac{c}{e} = \frac{2c}{e}. \end{aligned}$$

If $a < c/e$, then $a \Pr(X > a) = a < c/e$. If $c/e \le a < c$, then $a \Pr(X > a) = c/e$. If $a \ge c$, the product is zero. This proves (14). $\square$

Proof of Theorem 1(ii). Fix an arbitrary randomized policy π. Draw X according to Lemma 9, and present the terminal-jump sequence p^X. By Lemma 8, every realization belongs to $\mathcal{C}_T(g)$.

For each $t < T$, the observed history under every realization of X is the all-zero prefix. Therefore A_t is independent of X. The learner's reward from round t is

$$A_t \mathbf{1}\{A_t < X\}.$$

Conditioning on A_t and applying (14),

$$\mathbb{E}\big[A_t \mathbf{1}\{A_t < X\}\big] = \mathbb{E}\big[A_t \Pr(X > A_t \mid A_t)\big] \le \frac{c}{e}.$$

There is no reward from a quote submitted at time T. Hence

$$\mathbb{E}\left[\sum_{t=1}^{T} r_t^{p^X}(A_t)\right] \le \frac{c}{e}(T-1). \tag{15}$$

For a realized terminal value $X = x$, every quote $a < x$ posted before time T executes at time T. Thus the best fixed comparator value is

$$\sup_{a<x}(T-1)a = (T-1)x.$$

Averaging over X and using (14),

$$\mathbb{E}\left[\sup_{a\in[0,1]} \sum_{t=1}^{T} r_t^{p^X}(a)\right] = (T-1)\mathbb{E}[X] = \frac{2c}{e}(T-1). \tag{16}$$

Subtracting (15) from (16) shows that π has average regret at least $c(T-1)/e$ under this distribution over deterministic sequences. Therefore at least one realization p^x satisfies

$$R_T(\pi, p^x) \ge \frac{c}{e}(T-1).$$

Since this holds for every policy π and every such p^x lies in $\mathcal{C}_T(g)$, taking the infimum over policies proves (6). $\square$

6 Deriving the if-and-only-if statement

Proof of (7). Suppose first that $g(H) \to 0$. For $T \ge 2$, choose

$$H_T = \lfloor T^{1/3} \rfloor, \qquad K_T = T.$$

The upper bound (5) gives

$$\frac{\mathfrak{R}_T(g)}{T} \le \sqrt{\frac{(H_T+1)\log T}{2T}} + \frac{1}{T} + g(H_T).$$

Here $H_T \to \infty$ and $H_T \log T/T \to 0$, so all three terms converge to zero. Therefore $\mathfrak{R}_T(g) = o(T)$.

Conversely, if $g(H)$ does not converge to zero, then monotonicity implies that $c = \lim_H g(H) > 0$. By (6),

$$\liminf_{T\to\infty} \frac{\mathfrak{R}_T(g)}{T} \ge \frac{c}{e} > 0.$$

Thus the minimax regret is not sublinear. $\square$

7 Remarks on the scope of the characterization

Remark 10 (Why maximality matters). The iff statement is for the maximal class $\mathcal{C}_T(g)$ containing every sequence that satisfies the tail envelope. A smaller structured class can have very long delays and still be easy. For example, if the learner is told in advance that the unique possible sequence is $(0, \ldots, 0, 1)$, it can post quotes arbitrarily close to 1 from the first round. Thus a delay-tail condition alone cannot characterize every arbitrary subclass of deterministic sequences; it becomes necessary precisely because the maximal class includes all hard terminal-jump continuations allowed by the envelope.

Remark 11 (Interpretation of $\Gamma_{T,H}$). For the terminal-jump sequence p^x, Lemma 7 gives

$$\Gamma_{T,H}(p^x) = x(T - H - 1)_+.$$

The final $H + 1$ submission times do not count because their waiting times are at most H or there is no future round. The expression is nevertheless of order xT whenever $H = o(T)$.